

Section 1G. Two Random Variables

Victor M. Preciado, PhD
Dept of Electrical & Systems Engineering
University of Pennsylvania
preciado@seas.upenn.edu

Two Random Variables

Joint distributions for two r.v.'s X and Y are defined below:

- ▶ The **joint CDF** of two r.v.'s X and Y is defined as

$$F_{XY}(x, y) = \Pr(X \leq x, Y \leq y)$$

- ▶ The **joint PDF** is defined as $f_{XY}(x, y) = \partial^2 F_{XY}(x, y) / \partial x \partial y$. Hence, for a set $A \in \mathbb{R}^2$, we have that

$$\Pr((X, Y) \in A) = \iint_{(x, y) \in A} f_{XY}(x, y) dx dy$$

- ▶ The **joint PMF** of two discrete r.v.'s is defined by

$$p_{XY}(x, y) = \Pr(X = x, Y = y)$$

Two Random Variables: Marginalization

Marginal distributions are related to joint distributions as indicated below:

- ▶ The **marginal CDF** of X , F_X , satisfies

$$F_X(x) = \lim_{y \rightarrow \infty} F_{XY}(x, y)$$

- ▶ The **marginal PDF** (or marginal density) of X , f_X , satisfies

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

- ▶ The **marginal PMF** is given by

$$p_X(x) = \sum_{y \in \mathcal{Y}} p_{XY}(x, y)$$

where $\mathcal{Y}$ is the set of values that Y can take.

Two Random Variables: Conditioning

The probability distribution of a r.v. Y given that you already know that X takes the exact value x is described using the conditional distribution.

- ▶ Assuming to discrete r.v.'s X and Y , the **conditional PMF** of Y given X is:

$$p_{Y|X}(y|x) = \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} = \frac{p_{XY}(x, y)}{p_X(x)}$$

- ▶ Assuming to continuous r.v.'s X and Y , the **conditional PDF** of Y given X is:

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

Two Random Variables: Independence

- ▶ **Independence:** Two r.v.'s are independent if “knowing” the value of one variable does not affect the conditional probability distribution of the other variable.
 - ▶ Two discrete r.v.'s are independent if for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, we have

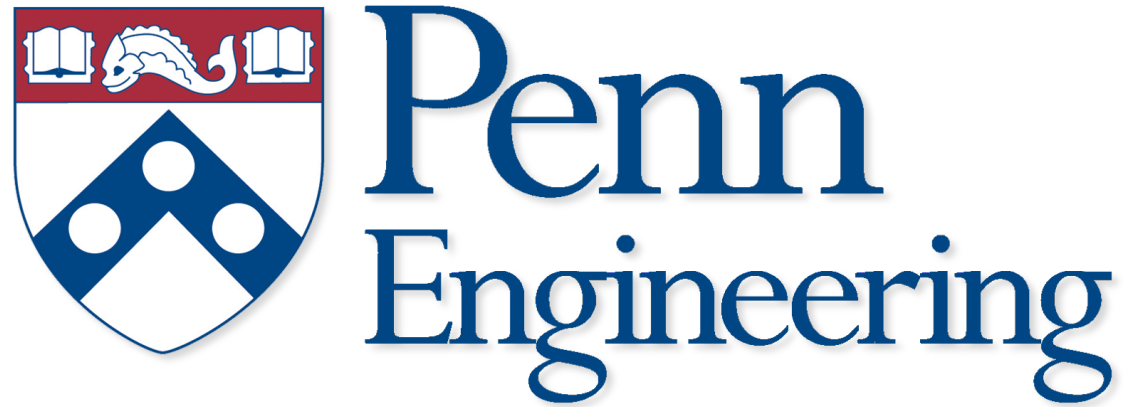
$$p_{XY}(x, y) = p_X(x) p_Y(y)$$

This is equivalent to $p_{X|Y}(x|y) = p_X(x)$ and $p_{Y|X}(y|x) = p_Y(y)$

- ▶ Two continuous r.v.'s are independent if for all $x, y \in \mathbb{R}$

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

This is equivalent to $f_{X|Y}(x|y) = f_X(x)$ and $f_{Y|X}(y|x) = f_Y(y)$



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